MARUTINOVATECH | @datawithdeepankar

Don't run.
Just don't stop.

70 DAYS
Data Analytics Roadmap

Slow and consistent beats fast and burnt out

One small step a day for 70 days. Each day tells you what to learn, gives you a free resource and a task, and asks one interview question. Tick it off, keep your streak alive, and let consistency do the heavy lifting.

Excel | SQL Server | Statistics | Power BI | Python + Pandas | Portfolio | Interviews

1-2
hours a day. That is all.

5
portfolio projects incl. a capstone

9
built-in review and catch-up days

Curated by **Deepankar Pathak**
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START HERE

How to use this roadmap

The daily routine

1. **Learn** the topic from the main resource (20 to 30 min).
2. **Practise** the task under "Do today" (40 to 60 min). This is where you actually learn.
3. **Answer** the interview question aloud, as if a recruiter asked it.
4. **Save** your work in one folder. It becomes your portfolio.

Stack order: Excel > SQL Server > Statistics > Power BI > Python > Portfolio > Interviews. Each tool reuses the last one, so your SQL database from Day 13 feeds Power BI on Day 40 and Pandas on Day 58.

The rules

- 1 to 2 hours a day is enough. Slow is fine, skipping is not.
- Every 7th day is a **Review + Catch-up day**. Use it to fix gaps, not to rest.
- Use the "Also" resource only when the main one does not click.
- Every resource has a free option. SQL Server Developer edition and Power BI Desktop cost nothing.
- **Setup:** a Windows laptop (SSMS and Power BI Desktop are Windows only) and Microsoft 365 or Excel 2024, since older Excel has no XLOOKUP, FILTER or TEXTSPLIT.
- Some SQL practice sites run a different engine. If you see LIMIT there, write TOP in SQL Server.
- Honest note: 70 days builds a strong, job-ready foundation and portfolio. The job comes from applying consistently.

The 70 days at a glance

Phase	Days	Focus	What you will have at the end
Phase 1	1 to 12	**Foundations + Excel**	Think like an analyst and become fast in Excel.
Phase 2	13 to 30	**SQL Server**	The most asked skill in analyst interviews. Biggest phase on purpose.
Phase 3	31 to 38	**Statistics + Analytical Thinking**	Numbers that hold up when a manager asks 'are you sure?'
Phase 4	39 to 50	**Power BI**	From raw tables to a published, secure report.
Phase 5	51 to 58	**Python for Analysis**	Pandas for anything Excel and SQL find painful.
Phase 6	59 to 63	**Portfolio, Capstone + AI**	Proof of work that recruiters can open and click.
Phase 7	64 to 70	**Interviews + Job Hunt**	Convert skills into offers.

Your 5 portfolio projects

1 Excel sales dashboard
Day 11

2 SQL e-commerce analysis
Day 30

3 Power BI report on SQL Server
Day 50

4 Python EDA with SQL + API
Day 58

5 End-to-end capstone
Days 60 to 61

THE ONLY RULE

Slow is fine. Stopping is not.

Most people quit a roadmap in week two because they try to finish it in a hurry. This one is built the other way round: small daily steps, protected by a few simple rules.

01
Walk, don't run

1 to 2 hours a day. If a day takes you two days, that is fine. Order matters, speed does not.

02
The 20-minute minimum

Busy, tired or sick? Do only the interview question and one small task. The streak stays alive.

03
Never miss twice

Missing one day is life. Missing two days in a row is the start of quitting. Come back the next day.

04
Track the streak

Tick the DONE box on every day and colour your tracker. Seeing the chain grow keeps you going.

05
Review days are sacred

Every 7th day you catch up and revise. Nothing new. This is what makes the learning stick.

06
Learn in public

Share your tracker and projects on your story. Tag @datawithdeepankar so I can cheer you on.

MY COMMITMENT

I, _____, will show up for 70 days, one small step at a time.

Start date _____ Target finish _____ My daily time slot _____

Why I am doing this _____

"Nothing in the world can take the place of persistence."

Calvin Coolidge. Ray Kroc, who built McDonald's, kept these words framed in his office.

YOUR STREAK

70-day tracker

Colour a box every day you show up. Print it, or screenshot it and tick it in your phone's photo editor. Share it every Sunday.

WEEK 1	1	2	3	4	5	6	7 ●
WEEK 2	8	9	10	11	12	13	14
WEEK 3	15	16	17	18	19	20	21 ●
WEEK 4	22	23	24	25	26	27	28
WEEK 5	29	30	31	32	33	34	35 ●
WEEK 6	36	37	38	39	40	41	42
WEEK 7	43	44	45	46	47	48	49
WEEK 8	50 ●	51	52	53	54	55	56
WEEK 9	57	58	59	60	61	62	63
WEEK 10	64	65	66	67	68	69	70 ●

☐ Learning day ☐ Review + catch-up ☐ Portfolio project ● Milestone

Share your tracker on your story with **#70DaysWithDeepankar** and tag **@datawithdeepankar**. Every week I reshare a few streaks.

PHASE 1 | DAYS 1 TO 12

Foundations + Excel

Think like an analyst and become fast in Excel.

CONSISTENCY STORY

THE 1% RULE

James Clear's book Atomic Habits made a simple bit of maths famous. Get 1% better every day and after a year you are about 37 times better, because 1.01 multiplied by itself 365 times is roughly 37.8. Get 1% worse every day and you slide close to zero.

Today's lesson is small. That is the whole point.

WEEK 1 *Just show up. That is the whole goal this week.*

DAY 1 ☐ DONE

The Analyst Role

LEARN What a data analyst does all day, the analytics lifecycle (ask, prepare, clean, analyse, share, act), the four types of analytics (descriptive, diagnostic, predictive, prescriptive) and how an analyst differs from a data scientist and a data engineer.

RESOURCE Microsoft Learn: Data Analyst career path Also: Alex The Analyst (YouTube)

DO TODAY Write 5 business questions for a food-delivery app. For each, name the metric and the data you would need.

INTERVIEW *What does a data analyst actually do day to day?*

DAY 2 ☐ DONE

Data Fundamentals

LEARN Structured vs unstructured data, categorical vs numerical, dimensions vs measures, the grain of a table, primary and foreign keys, and common file formats (CSV, XLSX, JSON).

RESOURCE roadmap.sh: Data Analyst Roadmap Also: IBM: What is data analytics?

DO TODAY Download one Maven dataset. Label every column with its type, the table's grain and the business meaning.

INTERVIEW *What is the grain of a table and why does it matter?*

DAY 3 ☐ DONE

Excel Basics + Tables

LEARN Ranges vs Excel Tables (Ctrl+T), relative, absolute and mixed references, sort and filter, SUM, AVERAGE, COUNT, COUNTA, MIN, MAX and number formats.

RESOURCE Microsoft Excel Help & Learning Also: Kevin Stratvert (YouTube)

DO TODAY Build a 100-row sales table and calculate 5 KPIs (revenue, orders, average order value, max sale, min sale).

INTERVIEW *Relative vs absolute reference? Why convert a range into a Table?*

DAY 4

DONE

Excel Logic + Conditional Maths

LEARN IF, IFS, AND, OR, IFERROR, COUNTIFS, SUMIFS, AVERAGEIFS. Think in business rules first, syntax second.

RESOURCE Leila Gharani (YouTube) Also: Microsoft: Excel functions (A to Z)

DO TODAY Add target-achieved, grade and customer-segment columns to your sales table.

INTERVIEW *SUMIF vs SUMIFS? When can IFERROR hide a real problem?*

DAY 5

DONE

Excel Lookups + Dynamic Arrays

LEARN XLOOKUP first, then VLOOKUP and INDEX + MATCH. Exact vs approximate match, why #N/A appears, plus UNIQUE, FILTER and SORT.

RESOURCE Microsoft: XLOOKUP function Also: Kevin Stratvert (YouTube)

DO TODAY Join a customer sheet to an orders sheet. Plant 5 missing keys on purpose and handle them cleanly.

INTERVIEW *XLOOKUP vs VLOOKUP? Why does a lookup fail when the value looks identical?*

DAY 6

DONE

Excel Text + Dates

LEARN LEFT, RIGHT, MID, LEN, TRIM, PROPER, TEXTSPLIT, CONCAT, TEXT; YEAR, MONTH, EOMONTH, NETWORKDAYS, DATEDIF. Learn that dates are stored as numbers.

RESOURCE Microsoft: Excel functions (A to Z) Also: Leila Gharani (YouTube)

DO TODAY Clean messy names and IDs, then build Year, Month and Quarter columns.

INTERVIEW *Why does a date column sometimes sort in the wrong order?*

DAY 7

DONE

Review Day: Excel Formulas REVIEW + CATCH-UP

LEARN Revise Days 1 to 6. Catch up on anything you skipped. No new topics today.

RESOURCE ExcelIsFun (YouTube) Also: Microsoft Excel Help & Learning

DO TODAY Redo 15 formulas without notes and keep a mistake log. Explain one formula aloud as if to your manager.

INTERVIEW *Explain the difference between COUNT, COUNTA and COUNTIFS.*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

MILESTONE

WEEK 1 DONE

The hardest part was starting. You did it.

WEEK 2

Excel gets faster when you stop rushing it.

DAY 8

DONE

Data Cleaning + Power Query in Excel

LEARN Duplicates, blanks, inconsistent labels, data validation and conditional formatting. Then Power Query: import, clean, and refresh repeatable steps instead of fixing by hand.

RESOURCE Microsoft Excel Help & Learning Also: Chandoo (YouTube)

DO TODAY Clean a messy customer file in Power Query and write a 10-line cleaning log.

INTERVIEW *How do you handle duplicates? When are missing values meaningful?*

DAY 9 ☐ DONE

PivotTables

LEARN Rows, columns, values, grouping dates, % of total, slicers and PivotCharts. Use them to answer real business questions fast.

RESOURCE Microsoft: Create a PivotTable Also: ExcelIsFun (YouTube)

DO TODAY Revenue by month, region, category and salesperson. Find the top 3 and bottom 3 performers.

INTERVIEW *When is a PivotTable better than formulas?*

DAY 10 ☐ DONE

Charts + Choosing the Right Visual

LEARN Match the chart to the question: comparison, trend, distribution, relationship or composition. Avoid misleading axes and decide the KPI hierarchy.

RESOURCE Microsoft: Create charts from start to finish Also: Storytelling with Data

DO TODAY Make 5 different charts and write one line on why each chart fits.

INTERVIEW *Bar vs line chart? When is a pie chart a poor choice?*

DAY 11 ☐ DONE

PROJECT 1: Excel Sales Dashboard PORTFOLIO PROJECT

LEARN Build an executive dashboard: KPI cards, monthly trend, top products, region split and slicers on one screen.

RESOURCE Chandoo (YouTube) Also: Kenji Explains (YouTube)

DO TODAY Finish dashboard v1 plus a short note on assumptions and 5 insights.

INTERVIEW *Walk me through your dashboard and one design decision.*

DAY 12 ☐ DONE

Excel Case Study + Interview Day

LEARN End to end: inspect, clean, pivot, chart, explain. Focus on the insight, not on decoration.

RESOURCE Microsoft Excel Help & Learning Also: ExcelIsFun (YouTube)

DO TODAY Solve a 'sales dropped 20%' case in Excel and record 5 spoken answers.

INTERVIEW *Sales dropped 20% last month. What do you check first?*

PHASE 2 | DAYS 13 TO 30

SQL Server

The most asked skill in analyst interviews. Biggest phase on purpose.

CONSISTENCY STORY

THE MAN WHO MOVED A MOUNTAIN

In a Bihar village, Dashrath Manjhi lost his wife after she fell on a rocky ridge and could not reach a hospital in time. With only a hammer and chisel he cut a path through that ridge, working for about 22 years until 1982. A 55 km detour became a short walk. Bollywood told his story in Manjhi: The Mountain Man (2015).

SQL is your mountain. Chip at it one query a day.

DAY 13 ☐ DONE

SQL Server Setup + Databases

LEARN Install SQL Server Developer edition (free) and SSMS (Windows only). Databases, schemas, tables, data types (INT, DECIMAL, VARCHAR vs NVARCHAR, DATE), CREATE TABLE, INSERT, primary and foreign keys, DDL vs DML.

RESOURCE Microsoft Learn: Get Started Querying with T-SQL Also: Microsoft Learn: SQL Server Management Studio

DO TODAY Create a mini store database: Customers, Products, Orders, OrderItems. Insert 20 rows in each.

INTERVIEW *DDL vs DML? Primary key vs foreign key?*

DAY 14 ☐ DONE

Review Day: Excel Wrap-up + SQL Setup REVIEW + CATCH-UP

LEARN Revise Days 8 to 13 and finish pending Excel work. Make sure SQL Server and SSMS run without errors.

RESOURCE SQLBolt (interactive lessons) Also: Microsoft Excel Help & Learning

DO TODAY Polish Project 1 and write 10 practice INSERT statements.

INTERVIEW *Explain your Excel dashboard in 60 seconds.*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

WEEK 3 *SQL feels hard for everyone at first. Stay with it.*

DAY 15 ☐ DONE

SELECT, WHERE + NULLs

LEARN SELECT, DISTINCT, WHERE, AND, OR, NOT, IN, BETWEEN, LIKE, IS NULL. Learn the logical execution order: FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY.

RESOURCE SQLBolt (interactive lessons) Also: techTFQ (YouTube)

DO TODAY Write 20 filter queries. For 5 of them, write the execution order next to the query.

INTERVIEW *Why can't you use a SELECT alias inside WHERE?*

DAY 16 ☐ DONE

Sorting, TOP + CASE

LEARN ORDER BY, TOP and TOP WITH TIES, OFFSET FETCH, column aliases, and CASE WHEN to turn raw values into business buckets.

RESOURCE Microsoft Docs: SELECT (Transact-SQL) Also: HackerRank SQL

DO TODAY Create customer tiers and price bands using CASE. Return the 5 costliest products.

INTERVIEW *Can you ORDER BY an alias? Why does it work there but not in WHERE?*

DAY 17 ☐ DONE

Aggregation, GROUP BY + HAVING

LEARN COUNT, SUM, AVG, MIN, MAX, COUNT(DISTINCT). Think about the level you are grouping at. WHERE filters rows, HAVING filters groups.

RESOURCE SQLBolt (interactive lessons) Also: DataLemur SQL interview questions

DO TODAY Solve 15 GROUP BY and HAVING problems.

INTERVIEW *COUNT(*) vs COUNT(column) vs COUNT(DISTINCT column)? WHERE vs HAVING?*

DAY 18 ☐ DONE

JOINs I

LEARN INNER, LEFT, RIGHT and FULL JOIN, join keys, one-to-many relationships and why a join can multiply rows.

RESOURCE SQLBolt (interactive lessons) Also: Luke Barousse: SQL for Data Analytics

DO TODAY Write 15 multi-table queries. Note the row count before and after every join.

INTERVIEW *INNER vs LEFT JOIN? Why did my join double the revenue?*

DAY 19 ☐ DONE

JOINs II + Set Operators

LEARN Self join, CROSS JOIN, anti-joins (LEFT JOIN with IS NULL, NOT EXISTS), UNION vs UNION ALL, INTERSECT, EXCEPT.

RESOURCE HackerRank SQL Also: LeetCode Top SQL 50 (choose MS SQL Server)

DO TODAY Find customers with no orders in 3 different ways. Build an employee-manager self join.

INTERVIEW *UNION vs UNION ALL? Find customers who never ordered.*

DAY 20 ☐ DONE

SQL Functions + NULL Handling

LEARN LEN, TRIM, UPPER, SUBSTRING, CHARINDEX, REPLACE, CONCAT; ISNULL, COALESCE, NULLIF; CAST, CONVERT, TRY_CAST, ROUND.

RESOURCE techTFQ (YouTube) Also: HackerRank SQL

DO TODAY Clean a messy customer table in SQL: trim names, fix casing, fill missing cities, safely convert text amounts.

INTERVIEW *ISNULL vs COALESCE? Why use NULLIF in a division?*

DAY 21 ☐ DONE

Review Day: SQL Core REVIEW + CATCH-UP

LEARN Revise Days 15 to 20. Re-solve the questions you got wrong.

RESOURCE LeetCode Top SQL 50 (choose MS SQL Server) Also: DataLemur SQL interview questions

DO TODAY 20 mixed questions in 60 minutes. Group your mistakes by topic.

INTERVIEW *Explain the logical execution order of a query aloud.*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

MILESTONE
21-DAY HABIT UNLOCKED Three weeks of showing up. Learning is now part of your routine.

WEEK 4 *Three weeks in. This is turning into a habit.*

DAY 22 ☐ DONE

Dates in SQL Server

LEARN GETDATE, DATEPART, YEAR, MONTH, DATEADD, DATEDIFF, EOMONTH, DATETRUNC (SQL Server 2022+), FORMAT. Bucket data by month and quarter.

RESOURCE Microsoft Docs: Date and time functions (T-SQL) Also: DataLemur SQL interview questions

DO TODAY 10 date queries: monthly revenue, days to ship, this month vs last month.

INTERVIEW *How do you get the first day of the month? What are the DATEDIFF traps?*

DAY 23 ☐ DONE

Subqueries + EXISTS

LEARN Scalar, multi-row and table subqueries, correlated subqueries, IN vs EXISTS, and when a JOIN is clearer.

RESOURCE HackerRank SQL Also: techTFQ (YouTube)

DO TODAY Solve 10 subquery problems and rewrite 3 of them as joins.

INTERVIEW *What is a correlated subquery? IN vs EXISTS?*

DAY 24 ☐ DONE

CTEs, Views + Temp Tables

LEARN WITH clauses, chained CTEs, a first look at recursive CTEs, CREATE VIEW and #temp tables.

RESOURCE LeetCode Top SQL 50 (choose MS SQL Server) Also: DataLemur SQL interview questions

DO TODAY Rewrite 5 messy queries with CTEs and save one report query as a view.

INTERVIEW *CTE vs temp table vs view?*

DAY 25 ☐ DONE

Window Functions I: Ranking

LEARN OVER, PARTITION BY, ORDER BY; ROW_NUMBER, RANK, DENSE_RANK, NTILE. Top-N per group and removing duplicates.

RESOURCE Microsoft Learn: Write queries that use window functions Also: Luke Barousse: SQL for Data Analytics

DO TODAY Top 3 products per category. Remove duplicate rows using ROW_NUMBER.

INTERVIEW *RANK vs DENSE_RANK vs ROW_NUMBER?*

DAY 26 ☐ DONE

Window Functions II: Analytics

LEARN LAG, LEAD, running totals with SUM() OVER, moving averages with ROWS BETWEEN, FIRST_VALUE and % of total.

RESOURCE Microsoft Docs: OVER clause (T-SQL) Also: DataLemur SQL interview questions

DO TODAY Running revenue, 3-month moving average and month-on-month growth %.

INTERVIEW *How do you calculate MoM growth? ROWS vs RANGE?*

DAY 27 ☐ DONE

Business Metrics in SQL

LEARN Average order value, repeat rate, customer lifetime value, inactive customers, cohort retention, contribution % and Pareto (80/20).

RESOURCE DataLemur SQL interview questions Also: Maven Analytics Data Playground (free datasets)

DO TODAY Answer 8 business questions using SQL only, with a one-line insight for each.

INTERVIEW *How would you define and calculate a repeat customer?*

DAY 28

DONE

Review Day: Advanced SQL REVIEW + CATCH-UP

LEARN Revise Days 22 to 27. Window functions and dates need the most repetition.

RESOURCE LeetCode Top SQL 50 (choose MS SQL Server) Also: techTFQ (YouTube)

DO TODAY Redo every wrong question from this week. Explain 3 window queries aloud.

INTERVIEW *Top 2 earners per department: walk through your query.*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

WEEK 5 *Slow learners who keep going finish first.*

DAY 29

DONE

SQL Interview Sprint + Performance Basics

LEARN Timed mixed practice. Plus what every analyst should know: clustered vs non-clustered indexes, SARGable filters and a first look at execution plans.

RESOURCE LeetCode Top SQL 50 (choose MS SQL Server) Also: DataLemur SQL interview questions

DO TODAY Solve 10 questions in 60 minutes, then redo every miss.

INTERVIEW *What is an index? Why can a function on a column in WHERE slow a query?*

DAY 30

DONE

PROJECT 2: SQL E-commerce Analysis PORTFOLIO PROJECT

LEARN Revenue trend, AOV, repeat customers, top products, cohort retention and inactive customers on your own database or a Maven dataset.

RESOURCE roadmap.sh: Data Analyst Projects Also: Maven Analytics Data Playground (free datasets)

DO TODAY Publish the .sql file, a schema diagram, 8 insights and your recommendations.

INTERVIEW *Walk me through your most complex query and its edge cases.*

PHASE 3 | DAYS 31 TO 38

Statistics + Analytical Thinking

Numbers that hold up when a manager asks 'are you sure?'

CONSISTENCY STORY

TOYOTA AND KAIZEN

Toyota became one of the world's biggest carmakers on kaizen, a Japanese idea of continuous improvement. Instead of rare, dramatic changes, everyone makes small improvements every single day, and those small changes keep adding up.

Small improvements, repeated, beat big bursts.

DAY 31 ☐ DONE

Descriptive Statistics

LEARN Mean, median, mode, range, variance, standard deviation, percentiles, quartiles and IQR.

RESOURCE Khan Academy: Statistics & Probability Also: StatQuest (YouTube)

DO TODAY Calculate these for one dataset in Excel and in SQL, and interpret each number.

INTERVIEW *When is the median better than the mean?*

DAY 32 ☐ DONE

Distributions + Outliers

LEARN Normal vs skewed distributions, z-scores, the IQR rule, histograms and box plots. Learn what the shape is telling you.

RESOURCE StatQuest (YouTube) Also: Seeing Theory (interactive)

DO TODAY Plot a histogram, flag the outliers and decide keep or remove, with a reason.

INTERVIEW *How do outliers affect the mean and standard deviation?*

DAY 33 ☐ DONE

Probability + Sampling

LEARN Basic and conditional probability, population vs sample, sampling methods, sampling bias and the intuition behind the Central Limit Theorem.

RESOURCE Khan Academy: Statistics & Probability Also: OpenIntro Statistics (free book)

DO TODAY Design a sampling plan for a customer satisfaction survey.

INTERVIEW *What is sampling bias? Give a real example.*

DAY 34 ☐ DONE

Correlation + Regression

LEARN Correlation vs causation, confounders, simple linear regression, R-squared intuition, and trendlines in Excel.

RESOURCE StatQuest (YouTube) Also: Khan Academy: Statistics & Probability

DO TODAY Analyse ad spend vs sales. Explain the relationship and its limits.

INTERVIEW *Correlation vs causation? What does R-squared tell you?*

DAY 35 ☐ DONE

Review Day: Statistics REVIEW + CATCH-UP

LEARN Revise Days 31 to 34 and finish Project 2.

RESOURCE Seeing Theory (interactive) Also: Khan Academy: Statistics & Probability

DO TODAY Write a one-page stats cheat sheet in your own words.

INTERVIEW *Explain standard deviation to a non-technical manager.*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

MILESTONE
HALFWAY THERE 35 days down. Screenshot your tracker and share how far you have come.

WEEK 6 *Past halfway. Your streak matters more than your speed.*

DAY 36 ☐ DONE

Hypothesis Testing

LEARN Null and alternative hypotheses, p-values, significance level, confidence intervals, t-test, the idea of chi-square, and Type I vs Type II errors.

RESOURCE Khan Academy: Statistics & Probability Also: StatQuest (YouTube)

DO TODAY Write a test plan for 'the new checkout page increased conversion'.

INTERVIEW *What does a p-value actually mean?*

DAY 37 ☐ DONE

A/B Testing

LEARN Control vs treatment, randomisation, sample size, primary and guardrail metrics, practical vs statistical significance, and the peeking problem.

RESOURCE Optimizely: A/B testing guide Also: StatQuest (YouTube)

DO TODAY Design an A/B test for a landing-page change, including sample size and success metric.

INTERVIEW *What makes an A/B test trustworthy?*

DAY 38 ☐ DONE

Analytical Thinking + KPIs

LEARN Problem > metric > data > analysis > insight > recommendation. KPI trees, funnel metrics and the basics of guesstimates.

RESOURCE roadmap.sh: What does a Data Analyst do? Also: Alex The Analyst (YouTube)

DO TODAY Turn 5 vague requests into measurable analysis plans. Build one KPI tree.

INTERVIEW *Revenue fell 20%. How would you investigate?*

PHASE 4 | DAYS 39 TO 50

Power BI

From raw tables to a published, secure report.

CONSISTENCY STORY

FROM KNOW-IT-ALL TO LEARN-IT-ALL

When Satya Nadella became Microsoft's CEO in 2014, he pushed a culture change built on a growth mindset. His message to employees: stop trying to be know-it-alls and become learn-it-alls. The company that makes Power BI rebuilt itself on the habit of learning.

You don't need to know DAX yet. You need to keep learning it.

DAY 39 ☐ DONE

Power BI Foundations

LEARN Install Power BI Desktop. Desktop vs Service, reports, semantic models, dashboards, data sources and the end-to-end BI workflow.

RESOURCE Microsoft Learn: Data Analyst career path Also: Guy in a Cube (YouTube)

DO TODAY Import an Excel and a CSV file and build your first report page.

INTERVIEW *Power BI Desktop vs Service? Report vs dashboard?*

DAY 40 ☐ DONE

Power Query I

LEARN Connect to Excel, CSV and SQL Server. Data types, column profiling, remove, rename, filter, replace values, and the idea of query folding.

RESOURCE Microsoft Learn: Prepare data for analysis Also: Guy in a Cube (YouTube)

DO TODAY Connect to your SQL Server database from Day 13 and clean one table. Document every step.

INTERVIEW *Where should a transformation happen: SQL, Power Query or DAX?*

DAY 41 ☐ DONE

Power Query II

LEARN Merge, append, split columns, fill down, unpivot, conditional columns, parameters and the Folder connector.

RESOURCE Microsoft Learn: Prepare data for analysis Also: How to Power BI (YouTube)

DO TODAY Append 3 monthly files from one folder. Unpivot one wide table.

INTERVIEW *Merge vs append? Why would you unpivot?*

DAY 42 ☐ DONE

Review Day: Stats + Power Query REVIEW + CATCH-UP

LEARN Revise Days 36 to 41. Catch up on anything pending.

RESOURCE Guy in a Cube (YouTube) Also: Khan Academy: Statistics & Probability

DO TODAY Redo one Power Query cleaning from scratch in under 30 minutes.

INTERVIEW *Explain each Power Query step you applied and why.*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

WEEK 7 *DAX clicks on the third try, not the first. Be patient.*

DAY 43 ☐ DONE

Data Modeling + Star Schema

LEARN Fact vs dimension tables, relationships, cardinality, filter direction, star vs snowflake, and a proper date table (mark it as a date table).

RESOURCE Microsoft Learn: Model data with Power BI Also: SQLBI (data modeling articles)

DO TODAY Build a star schema with a date table. Screenshot the model view.

INTERVIEW *Why a star schema? Why avoid bi-directional filters?*

DAY 44

DONE

DAX Basics

LEARN Measures vs calculated columns; SUM, COUNTROWS, DISTINCTCOUNT, DIVIDE, and iterators like SUMX and AVERAGEX.

RESOURCE Microsoft Learn: Use DAX in Power BI Also: SQLBI: DAX Basics learning path

DO TODAY Create 10 measures: revenue, orders, customers, AOV, margin % and more.

INTERVIEW *Measure vs calculated column? SUM vs SUMX?*

DAY 45

DONE

CALCULATE + Filter Context

LEARN Row context vs filter context, CALCULATE, FILTER, ALL, REMOVEFILTERS and variables. Understand why a measure changes under filters.

RESOURCE SQLBI: DAX Basics learning path Also: Microsoft Learn: Use DAX in Power BI

DO TODAY Build revenue % of total, category share and filtered margin measures.

INTERVIEW *What does CALCULATE do?*

DAY 46

DONE

DAX Time Intelligence

LEARN Date table, TOTALYTD, DATESYTD, SAMEPERIODLASTYEAR, DATEADD, MTD and YoY %.

RESOURCE Microsoft Learn: Use DAX in Power BI Also: SQLBI: Time intelligence in Power BI

DO TODAY Build YTD, last-year and YoY % measures. Check one number by hand with a SQL query.

INTERVIEW *Why do time intelligence functions need a date table?*

DAY 47

DONE

Visuals, Interactivity + Storytelling

LEARN Cards, matrix, bar, line, scatter and maps; slicers, drill-through, tooltip pages and bookmarks; layout, whitespace and restrained colour.

RESOURCE Microsoft: Power BI learning directory Also: Storytelling with Data

DO TODAY Redesign a cluttered page into a one-page executive view with drill-through.

INTERVIEW *What makes a dashboard actionable?*

DAY 48

DONE

Power BI Service + Security

LEARN Publish, workspaces, scheduled refresh, gateways, sharing and apps, Import vs DirectQuery, and row-level security (RLS).

RESOURCE Microsoft Docs: Row-level security in Power BI Also: Guy in a Cube (YouTube)

DO TODAY Set RLS by region and test it with 'View as' in Desktop. Publishing needs a work or school email; Gmail will not work.

INTERVIEW *Import vs DirectQuery? How does RLS work?*

DAY **49** DONE

Review Day: Modeling + DAX REVIEW + CATCH-UP

LEARN Revise Days 43 to 48. DAX sticks only with repetition.

RESOURCE SQLBI: DAX Basics learning path Also: Guy in a Cube (YouTube)

DO TODAY Rewrite 10 measures from memory and verify each against a SQL query.

INTERVIEW *Explain filter context with an example.*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

WEEK 8 *New language, same patience. One cell at a time.*

DAY **50** DONE

PROJECT 3: Power BI Report on SQL Server PORTFOLIO PROJECT

LEARN A 3-page report (overview, customers, products) on your SQL Server data: star schema, 10+ measures, drill-through and RLS.

RESOURCE Microsoft Learn: Data Analyst career path Also: SQLBI: Learning paths

DO TODAY Publish screenshots, the PBIX, a model diagram and 8 insights.

INTERVIEW *Walk me through your model and two of your measures.*

MILESTONE
50 DAYS STRONG Excel, SQL, statistics and Power BI. Look at how much you know now.

PHASE 5 | DAYS 51 TO 58

Python for Analysis

Pandas for anything Excel and SQL find painful.

CONSISTENCY STORY

PYTHON WAS A HOLIDAY SIDE PROJECT

In December 1989, Guido van Rossum used his Christmas holidays at CWI in Amsterdam to start writing a new language as a side project. He named it after the comedy show Monty Python's Flying Circus and released it in 1991. Today it powers data work across the world.

Big things start as small, quiet projects.

DAY **51** DONE

Python Setup + Basics

LEARN Jupyter or Google Colab. Variables, data types, strings, lists, dictionaries, conditions and loops. Only the Python an analyst needs.

RESOURCE Kaggle Learn: Python Also: freeCodeCamp: Python course

DO TODAY Finish 10 small exercises in a notebook.

INTERVIEW *List vs tuple? What is a dictionary?*

DAY 52 ☐ DONE

Functions + NumPy

LEARN Functions, list comprehensions, basic error handling; NumPy arrays, indexing and vectorised maths.

RESOURCE NumPy: Learn Also: Kaggle Learn: Python

DO TODAY Write 5 functions and perform 10 array operations.

INTERVIEW *Why is NumPy faster than a Python loop?*

DAY 53 ☐ DONE

Pandas I: Exploring Data

LEARN Series and DataFrames, read_csv, read_excel, head, info, describe, selecting columns, loc vs iloc and boolean filtering.

RESOURCE Kaggle Learn: Pandas Also: pandas: Getting started

DO TODAY Load a CSV and answer 10 questions about it.

INTERVIEW *loc vs iloc? Series vs DataFrame?*

DAY 54 ☐ DONE

Pandas II: Cleaning

LEARN Missing values, dtypes, duplicates, string methods, to_datetime, fillna, dropna and renaming.

RESOURCE Kaggle Learn: Data Cleaning Also: Kaggle Learn: Pandas

DO TODAY Clean the same messy file you cleaned in Excel. Compare the effort.

INTERVIEW *How do you handle missing values?*

DAY 55 ☐ DONE

Pandas III: Aggregating + Combining

LEARN groupby, agg, pivot_table, merge, concat, sorting and resampling dates.

RESOURCE pandas: User guide Also: Kaggle Learn: Pandas

DO TODAY Recreate your SQL project's monthly revenue and top products in Pandas.

INTERVIEW *merge vs concat? groupby vs pivot_table?*

DAY 56 ☐ DONE

Review Day: Python REVIEW + CATCH-UP

LEARN Revise Days 51 to 55.

RESOURCE Kaggle Learn: Pandas Also: pandas: User guide

DO TODAY Build a 'SQL to Pandas' cheat sheet (WHERE = boolean filter, GROUP BY = groupby, JOIN = merge).

INTERVIEW *How would you write a SQL GROUP BY in Pandas?*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

WEEK 9 *Your work is starting to speak for you.*

DAY 57 ☐ DONE

Python Visualisation

LEARN Matplotlib and Seaborn: bar, line, histogram, box, scatter and heatmap. Write titles that state the insight.

RESOURCE Kaggle Learn: Data Visualization Also: Matplotlib tutorials

DO TODAY Make 6 charts with one insight each.

INTERVIEW *Histogram vs box plot: when do you use each?*

DAY 58 ☐ DONE

PROJECT 4: Python EDA (SQL + API) PORTFOLIO PROJECT

LEARN EDA workflow, reading a SQL Server table into Pandas, and pulling JSON from a public API with requests. Use Jupyter on your laptop here, since Colab cannot reach your local SQL Server.

RESOURCE Kaggle Learn (all courses) Also: Requests library docs

DO TODAY Publish a notebook with clean markdown, 8 insights and one limitation.

INTERVIEW *What is an API? Walk me through your EDA steps.*

PHASE 6 | DAYS 59 TO 63

Portfolio, Capstone + AI

Proof of work that recruiters can open and click.

CONSISTENCY STORY

INFOSYS STARTED WITH RS 10,000

In 1981, seven young engineers started Infosys in Pune. The starting money was Rs 10,000 that N. R. Narayana Murthy borrowed from his wife, Sudha Murty. Decades of steady work later, it is one of India's largest IT companies.

Your portfolio starts small too. Publish it anyway.

DAY 59 ☐ DONE

GitHub + Portfolio Setup

LEARN One repo per project. README structure: problem, data, approach, insights, recommendations, screenshots. Profile README and LinkedIn Featured section.

RESOURCE GitHub Docs: Hello World Also: roadmap.sh: Data Analyst Projects

DO TODAY Push all 4 projects with proper READMEs.

INTERVIEW *What business decision does your project support?*

DAY 60 ☐ DONE

CAPSTONE Part 1: Data + KPIs PORTFOLIO PROJECT

LEARN Pick one domain (banking, e-commerce or food delivery). Load raw data into SQL Server, clean it with SQL or Python and define 6 KPIs.

RESOURCE Maven Analytics Data Playground (free datasets) Also: roadmap.sh: Data Analyst Projects

DO TODAY Clean tables plus a one-page KPI definition document.

INTERVIEW *How did you choose your KPIs?*

DAY 61 ☐ DONE

CAPSTONE Part 2: Report + Presentation PORTFOLIO PROJECT

LEARN Build the Power BI report on your SQL tables and a 5-slide summary of findings and recommendations.

RESOURCE Microsoft Learn: Data Analyst career path Also: Storytelling with Data

DO TODAY Record a 5-minute walkthrough video of the capstone.

INTERVIEW *Present your project in 3 minutes.*

DAY 62 ☐ DONE

AI for Analysts

LEARN Copilot in Excel and Power BI, using AI assistants to draft SQL and DAX, checking AI output against the data, and never pasting confidential company data. Copilot needs a paid licence; free AI assistants are fine for practice.

RESOURCE Microsoft: Get started with Copilot in Excel Also: Microsoft: Power BI learning directory

DO TODAY Ask an AI to write 5 SQL queries for your project. Test each one and note which were wrong and why.

INTERVIEW *How do you use AI at work, and how do you verify it?*

DAY 63 ☐ DONE

Review Day: Portfolio Check REVIEW + CATCH-UP

LEARN Revise everything you built. Ask a friend or mentor to review your GitHub and LinkedIn.

RESOURCE roadmap.sh: Data Analyst Projects Also: Alex The Analyst (YouTube)

DO TODAY Fix 5 issues found in the review.

INTERVIEW *Which of your projects are you proudest of, and why?*

CHECK-IN What clicked this week? What still feels shaky? Write one small win and share it on your story.

PHASE 7 | DAYS 64 TO 70

Interviews + Job Hunt

Convert skills into offers.

CONSISTENCY STORY

THE FOUNDER STARTED AT 52

Ray Kroc was 52 and selling milkshake mixers when he visited the McDonald brothers' restaurant in 1954. After years of ordinary sales jobs, he went on to build McDonald's into a global chain, a journey shown in the film The Founder (2016). A framed note about persistence hung in his office.

It is never too late to start. Keep applying.

WEEK 10 *Finish the walk. Then keep walking into applications.*

DAY 64 ☐ DONE

SQL Interview Sprint

LEARN Classic patterns: second highest salary, duplicates, running totals, top-N per group, consecutive days and gaps.

RESOURCE LeetCode Top SQL 50 (choose MS SQL Server) Also: DataLemur SQL interview questions

DO TODAY Solve 15 questions timed and redo every miss the next morning.

INTERVIEW *Find users who logged in 3 days in a row.*

DAY 65 ☐ DONE

Excel + Power BI Interview

LEARN Rapid revision: lookups, Pivots, Power Query, star schema, DAX, RLS and dashboard design.

RESOURCE Microsoft Learn: Data Analyst career path Also: Guy in a Cube (YouTube)

DO TODAY Answer 25 technical questions aloud and record yourself.

INTERVIEW *XLOOKUP? Star schema? Measure vs column? CALCULATE?*

DAY 66 ☐ DONE

Stats, Case Studies + Guesstimates

LEARN Revenue drop, conversion drop, profit decline, churn spike and experiment design. Structured guesstimates.

RESOURCE Khan Academy: Statistics & Probability Also: StatQuest (YouTube)

DO TODAY Complete 4 case studies and 2 guesstimates using a clear structure.

INTERVIEW *How many cups of tea are sold in Jaipur every day?*

DAY 67 ☐ DONE

Resume + LinkedIn

LEARN A one-page, ATS-friendly resume, project bullets that show impact, and a LinkedIn headline, About and Featured section.

RESOURCE roadmap.sh: Data Analyst Career Path Also: Alex The Analyst (YouTube)

DO TODAY Finish resume v1 and update LinkedIn.

INTERVIEW *Tell me about yourself (in 60 to 90 seconds).*

DAY **68** DONE

Mock Interview

LEARN Technical round, project walkthrough and behavioural questions using the STAR method.

RESOURCE Alex The Analyst: Free Data Analyst Bootcamp Also: Alex The Analyst (YouTube)

DO TODAY Record a 30-minute mock interview and fix your 5 weakest answers.

INTERVIEW *Tell me about a mistake you found in data.*

DAY **69** DONE

Job Hunt System

LEARN Target roles (Data Analyst, MIS Analyst, Business Analyst, Reporting Analyst, Power BI Developer), job portals, referrals and outreach messages.

RESOURCE LinkedIn Jobs: Data Analyst Also: Naukri.com

DO TODAY Build a 30-application tracker and send 10 referral requests.

INTERVIEW *Why should we hire you?*

DAY **70** DONE

Final Self-Test + Next 90 Days

LEARN Take a full self-test across every phase. Then choose your next path: PL-300 certification, data engineering (Azure, Microsoft Fabric) or advanced Python.

RESOURCE Microsoft Learn: Data Analyst career path Also: Alex The Analyst: Free Data Analyst Bootcamp

DO TODAY Write your 90-day plan and keep applying every single day.

INTERVIEW *What will you learn next, and why?*

MILESTONE **YOU FINISHED** 70 days of consistency. That habit will carry your whole career.

INTERVIEW PREP

Question bank: what to expect

Excel

- XLOOKUP vs VLOOKUP?
- Relative vs absolute reference?
- When would you use a PivotTable?
- How do you find and remove duplicates?
- What is Power Query and why use it?

SQL Server

- INNER vs LEFT JOIN?
- WHERE vs HAVING?
- COUNT(*) vs COUNT(column)?
- RANK vs DENSE_RANK vs ROW_NUMBER?
- Find customers with no orders.
- Second highest salary per department.
- Calculate MoM growth.
- CTE vs temp table vs view?
- UNION vs UNION ALL?
- What is an index?

Statistics

- Mean vs median?
- How do outliers affect analysis?
- Correlation vs causation?
- What does a p-value mean?
- Type I vs Type II error?
- How would you design an A/B test?

Power BI

- Power Query vs DAX?
- Measure vs calculated column?
- What is a star schema?
- What is filter context?
- What does CALCULATE do?
- Import vs DirectQuery?
- How does row-level security work?

Python / Pandas

- loc vs iloc?
- Series vs DataFrame?
- merge vs concat?
- How do you handle missing values?
- How does groupby work?

Case studies

- Revenue fell 20%. How do you investigate?
- Conversion fell but traffic stayed the same. What do you check?
- Revenue went up but profit fell. Why?
- A dashboard shows a sudden spike. How do you validate it?

Behavioural

- Tell me about yourself.
- Walk me through a project.
- What was the hardest part of your analysis?
- Tell me about a mistake you found in data.
- How did your analysis lead to a decision?

BEFORE YOU APPLY

Portfolio checklist

- [ ] 5 projects: Excel, SQL, Power BI, Python and one end-to-end capstone.
- [ ] Every project shows problem, data, cleaning, analysis, visuals, insights and recommendations.
- [ ] Each GitHub repo has a README with screenshots, not just code.
- [ ] Power BI project includes the model diagram and your key DAX measures.
- [ ] SQL project includes the .sql file and a schema diagram.
- [ ] Resume links straight to the projects, one page, ATS friendly.
- [ ] LinkedIn headline and Featured section show your projects.
- [ ] A 3-minute spoken walkthrough ready for every project.

ALL FREE

Resource library

Every link in this roadmap in one place. Tap any name to open it.

Roadmaps + careers

roadmap.sh: Data Analyst Roadmap
roadmap.sh/data-analyst

roadmap.sh: What does a Data Analyst do?
roadmap.sh/data-analyst/what-does-a-data-analyst-do

roadmap.sh: Data Analyst Projects
roadmap.sh/data-analyst/projects

roadmap.sh: Data Analyst Career Path
roadmap.sh/data-analyst/career-path

Microsoft Learn: Data Analyst career path
learn.microsoft.com/en-us/training/career-paths/data-analyst

Alex The Analyst (YouTube)
www.youtube.com/@AlexTheAnalyst

Alex The Analyst: Free Data Analyst Bootcamp
www.youtube.com/watch?v=cnjhHZNJEDk

IBM: What is data analytics?
www.ibm.com/think/topics/data-analytics

Excel

Microsoft Excel Help & Learning
support.microsoft.com/excel

Microsoft: Excel functions (A to Z)
support.microsoft.com/office/excel-functions-alphabetical-b3944572-255d-4efb-bb96-c6d90033e188

Microsoft: XLOOKUP function
support.microsoft.com/office/xlookup-function-b7fd680e-733b-4d86-88b3-0be7d2c2f2d1

Microsoft: Create a PivotTable
support.microsoft.com/office/create-a-pivottable-to-analyze-worksheet-data-a9a84538-bfe9-40a9-a8e9-f99134456576

Microsoft: Create charts from start to finish
support.microsoft.com/en-us/excel/creating-charts-from-start-to-finish

Microsoft: Get started with Copilot in Excel
support.microsoft.com/en-us/copilot-excel

ExcelIsFun (YouTube)
www.youtube.com/@excelisfun

Leila Gharani (YouTube)
www.youtube.com/@LeilaGharani

Kevin Stratvert (YouTube)
www.youtube.com/@KevinStratvert

Chandoo (YouTube)
www.youtube.com/@chandoo_

Kenji Explains (YouTube)
www.youtube.com/@KenjiExplains

SQL Server

SQL Server downloads (Developer edition is free)
www.microsoft.com/en-us/sql-server/sql-server-downloads

Microsoft Learn: SQL Server Management Studio
learn.microsoft.com/sql/ssms

Microsoft Learn: Get Started Querying with T-SQL
learn.microsoft.com/en-us/training/paths/get-started-querying-with-transact-sql/

Microsoft Docs: SELECT (Transact-SQL)
learn.microsoft.com/en-us/sql/t-sql/queries/select-transact-sql

Microsoft Docs: Date and time functions (T-SQL)
learn.microsoft.com/en-us/sql/t-sql/functions/date-and-time-data-types-and-functions-transact-sql

Microsoft Learn: Write queries that use window functions
learn.microsoft.com/en-us/training/modules/write-queries-that-use-window-functions/

Microsoft Docs: OVER clause (T-SQL)
learn.microsoft.com/en-us/sql/t-sql/queries/select-over-clause-transact-sql

SQLBolt (interactive lessons)
sqlbolt.com/

HackerRank SQL
www.hackerrank.com/domains/sql

LeetCode Top SQL 50 (choose MS SQL Server)
leetcode.com/studyplan/top-sql-50/

DataLemur SQL interview questions
datalemur.com/questions

techTFQ (YouTube)
www.youtube.com/@techTFQ

Luke Barousse: SQL for Data Analytics
www.lukebarousse.com/sql

Statistics

Khan Academy: Statistics & Probability
www.khanacademy.org/math/statistics-probability

StatQuest (YouTube)
www.youtube.com/@statquest

Seeing Theory (interactive)
seeing-theory.brown.edu/

OpenIntro Statistics (free book)
www.openintro.org/book/os/

Optimizely: A/B testing guide
www.optimizely.com/optimization-glossary/ab-testing/

Power BI

Microsoft: Power BI learning directory
learn.microsoft.com/en-us/power-bi/fundamentals/power-bi-learning-path-directory

Microsoft Learn: Prepare data for analysis
learn.microsoft.com/en-gb/training/paths/prepare-data-power-bi/

Microsoft Learn: Model data with Power BI
learn.microsoft.com/en-us/training/paths/model-data-power-bi/

Microsoft Learn: Use DAX in Power BI
learn.microsoft.com/en-gb/training/paths/dax-power-bi/

Microsoft Docs: Row-level security in Power BI
learn.microsoft.com/power-bi/enterprise/service-admin-rls

SQLBI (data modeling articles)
www.sqlbi.com/

SQLBI: DAX Basics learning path
www.sqlbi.com/learning-paths/dax-basics/

SQLBI: Time intelligence in Power BI
www.sqlbi.com/articles/time-intelligence-in-power-bi/

Guy in a Cube (YouTube)
www.youtube.com/@GuyInACube

How to Power BI (YouTube)
www.youtube.com/@HowtoPowerBI

Storytelling with Data
www.storytellingwithdata.com/

Python

Kaggle Learn: Python
www.kaggle.com/learn/python

freeCodeCamp: Python course
www.freecodecamp.org/learn/scientific-computing-with-python/

NumPy: Learn
numpy.org/learn/

Kaggle Learn: Pandas
www.kaggle.com/learn/pandas

pandas: Getting started
pandas.pydata.org/docs/getting_started/index.html

pandas: User guide
pandas.pydata.org/docs/user_guide/index.html

Kaggle Learn: Data Cleaning
www.kaggle.com/learn/data-cleaning

Kaggle Learn: Data Visualization
www.kaggle.com/learn/data-visualization

Matplotlib tutorials
matplotlib.org/stable/tutorials/index.html

Requests library docs
requests.readthedocs.io/en/latest/

Portfolio + jobs

GitHub Docs: Hello World
docs.github.com/en/get-started/start-your-journey/hello-world

Maven Analytics Data Playground (free datasets)
mavenanalytics.io/data-playground

LinkedIn Jobs: Data Analyst
www.linkedin.com/jobs/data-analyst-jobs/

Naukri.com
www.naukri.com/

Roz thoda, par roz.

Keep building.
Keep applying.
Keep learning.

Save this roadmap, start Day 1 today and share your progress. Tag **@datawithdeepankar** when you finish a project and I will check it out.

INSTAGRAM

@datawithdeepankar

Daily SQL, Power BI and interview content.

YOUTUBE

Deepankar Pathak

Full-length lessons and walkthroughs.

FREE GUIDANCE SESSION

Book a 1:1 call

Stuck, or not sure where to start? Book a free call.

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Curated by
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